

United International University
Department of Computer Science and Engineering

CSE-3313: Computer Architecture

Final Examination: Fall 2025

Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from
the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. (a) Modify and annotate the block diagram for the single-cycle data path and **write the control unit values** for this instruction: '**cs \$s0, \$s1, \$s2**'. This is a R type instruction. The task of the instruction is to read the value stored at register \$s1, and multiply it by 8. The product of the multiplication should be stored at \$s0. **Afterwards**, the Program Counter (PC) will update itself with the address stored at the register \$s2. The value of \$s0, \$s1, \$s2 are stored in register rd, rs, and rt in the machine code translation. The machine code format for the instruction is given below in table 1: [5]

opcode	rs	rt	rd	shamt	function code
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits

Table 1: Machine code format of instruction cs \$s0, \$s1, \$s2

- (b) Annotate the block diagram for the single-cycle data path and **write the control unit values** for this instruction: '**ds \$s0, \$s1, 1024**'. The task of this instruction is to add the value given in \$s1 and offset and store it at \$s0. The value of the register number of \$s0 and \$s1 is stored at rt and rs, and the constant value 1024 is stored at the offset. The machine code format for ds is given in table 2: [5]

op	rs	rt	offset
6 bits	5 bits	5 bits	16 bits

Table 2: Machine code format of instruction ds

- (c) Modify and annotate the block diagram for a single-cycle data path, and write the control unit values of the following instruction so that it can execute the following custom J type instruction '**js 1024**'. Assuming the value 1024 is stored at the constant field, the task of this instruction is to read the value stored at the constant field, and update the value of the Program Counter using the following formula: **PC = PC + 1024 * 4**. The machine code format for the instruction is given below in table 3. [5]

op	constant
6 bits	26 bits

Table 3: Machine code format of instruction js

- (d) i. Assume we are executing a branch instruction (beq) at line number 440. If the condition is satisfied, the instruction at address 1004 will be executed. Write the branch instruction and briefly explain your answer. [3]
- ii. What are the PC values in the Single-Cycle Datapath after executing the instructions 'slt and jr'? [2]
2. Consider a processor that goes through the following seven stages in Table 4. while executing an instruction. Assume that when one instruction is using the memory stage to read or write, no other instruction can use the memory for write or another read in the same cycle.
- (a) Identify all pipeline hazards present in the instruction sequence. For each hazard, specify the type of hazard and the pair(s) of instruction involved. Resolve the hazards by reducing cycles using pipeline's hazard handling policy and compute the total execution time. Provide a timing diagram. [6]

```
1 addi $s0, $zero, 1024
2 sw $t2, 0($s0)
3 lw $t3, 4($s0)
4 add $t1, $t2, $t3
5 sub $t4, $s6, $s7
6 sw $t4, 8($s0)
```

Instruction Fetch (IF)	100 ns
Instruction Decode (ID)	120 ns
Register Read (RR)	120 ns
Execution (EXE)	220 ns
Memory Read (MEMR)	160 ns
Memory Write (MEMW)	170 ns
Write Back (WB)	190 ns

Table 4: Seven stages pipeline

- (b) Rearrange the instructions to minimize the stalls in (a) without changing the program semantics. Draw the pipeline timing diagram for the scheduled sequence and compute the total execution time. Explain how code scheduling reduces stalls in this example. [6]

- (c) Calculate the total execution time of the following code using single-cycle processor. [4]

```

1      addi $s1, $zero, 16
2      addi $s2, $zero, 1
3      addi $s3, $zero, 0
4  LOOP:
5      slt $t0, $s1, $s2
6      beq $t0, $zero, END
7      add $t1, $t0, $s0
8      sw  $t1, 0($s3)
9      sll $s2, $s2, 1
10     srl $s1, $s1, 1
11  END:
12     sw  $s3, 0($t4)

```

- (d) Analyze the pipeline execution for the following code when the branch is taken and not taken. Indicate stalls (if any), and compute the total number of clock cycles for each scenario. [4]

```

1
2      100: beq  $t0, $t1, 112
3      104: add  $t1, $t0, $s0
4      108: sw   $t3, 0($t1)
5      112: sll  $s2, $t3, 1

```